

AI/ML-Driven Digital Terrain Modelling and Optimized Drainage Network Design for Flood-Prone Village Abadi Areas

MoPR Geospatial Intelligence Challenge - Ministry of Panchayati Raj

IIT Tirupati Navavishkar I-Hub Foundation (IITNiF) — National
Informatics Centre (NIC)

Team DTM Drainage AI

Submission Date: April 2026

June 27, 2026

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1 Abstract

This report presents a comprehensive AI/ML-driven solution for Digital Terrain Model (DTM) generation and optimized drainage network design using drone-captured LiDAR point cloud data. The proposed system addresses the challenge of delineating natural surface-water flow paths, predicting waterlogging hotspots, and designing cost-effective drainage infrastructure for densely populated village (abadi) areas. The pipeline integrates classical geospatial algorithms with machine learning classifiers (Random Forest, XGBoost) and deep learning (PointNet) to achieve robust terrain characterization. Our approach produces OGC-compliant outputs including Cloud-Optimized GeoTIFFs (COG) and GeoPackages (GPKG), suitable for direct integration with GIS platforms deployed by Panchayati Raj institutions.

Keywords: Digital Terrain Model, LiDAR, Point Cloud Classification, Drainage Network Design, Machine Learning, XGBoost, Hydrological Analysis, Waterlogging Prediction, OGC Standards.

2 Introduction

2.1 Background and Motivation

The SVAMITVA (Survey of Villages and Mapping using Imagery and Technology for Abadi) scheme has enabled large-scale generation of high-resolution aerial imagery and LiDAR point cloud data for rural Indian villages. However, transforming this raw point cloud data into actionable geospatial intelligence for village planning remains a significant challenge. The absence of automated tools for Digital Terrain Model (DTM) creation and drainage network design forces engineers to rely on manual processing, which is time-consuming, error-prone, and inadequately scaled for the 6.5 lakh villages in India.

Waterlogging and inadequate drainage in abadi areas result in annual flood losses estimated at Rs. 15,000 crores across the country. Traditional drainage design relies on field surveys and approximate calculations, often leading to undersized or misaligned channels. The integration of AI/ML techniques offers a transformative approach: automated terrain analysis, predictive waterlogging modelling, and optimized hydraulic design—all within a reproducible, scalable pipeline.

2.2 Problem Statement

The objective of this research is to develop a data-driven DTM creation pipeline using drone point cloud datasets, combined with AI/ML-based waterlogging prediction and optimized drainage network design for densely inhabited village (abadi) areas. Specifically, the system must:

1. **Classify ground points** from raw LiDAR data using a hybrid morphological-ML approach;
2. **Generate a high-resolution DTM** (0.5 m) suitable for hydrological analysis;
3. **Delineate natural surface-water flow paths** and identify low-lying zones;
4. **Predict waterlogging hotspots** using terrain-derived ML features;
5. **Design an optimized drainage network** with hydraulic sizing and cost estimation.

The pipeline must produce GIS-ready outputs compliant with OGC standards (COG, GPKG, LAS 1.4).

2.3 Objectives

- Develop an automated three-stage ground classification pipeline combining PDAL-based morphological filters (SMRF, CSF) with Random Forest refinement;
- Implement IDW interpolation with Gaussian smoothing for DTM generation from classified ground points;
- Perform hydrological analysis using pysheds to compute flow direction, accumulation, and Topographic Wetness Index (TWI);
- Train an XGBoost classifier for waterlogging prediction using terrain features;
- Design a minimum spanning tree (MST) optimized drainage network using Manning's equation for hydraulic sizing;

- Generate all outputs in OGC-compliant formats.

2.4 Scope and Limitations

This work focuses on Gujarat state datasets (SVAMITVA) as proof-of-concept. The methodology is transferable to other states with similar LiDAR coverage. Key limitations include:

- Lack of ground-truth validation data for waterlogging events;
- Dependency on QGIS-bundled PDAL binaries for Windows compatibility;
- Computational requirements for processing 160M+ point datasets.

2.5 OGC Standards Compliance

This pipeline adheres to Open Geospatial Consortium (OGC) standards for interoperability and GIS integration:

Output	Format	Standard	Use Case
DTM Raster	Cloud-Optimized GeoTIFF	OGC COG	Efficient streaming, cloud storage
Hydrology Rasters	GeoTIFF	OGC GeoTIFF	Universal raster format
Drainage Vector	GeoPackage	OGC GPKG	Native GIS exchange
Point Cloud	LAS 1.4	ASPRS/LASF	LiDAR industry standard

The outputs are directly compatible with QGIS, ArcGIS, and other OGC-compliant GIS platforms deployed by state PRED departments and NIC.

The outputs are directly compatible with QGIS, ArcGIS, and other OGC-compliant GIS platforms deployed by state PRED departments and NIC.

2.6 Competitive Analysis

Existing tools for DTM generation and drainage design typically require manual intervention or expensive commercial licenses. Our solution provides a fully automated, open-source alternative:

Tool	Auto	ML	Drainage	Cost	Format
PDAL + GRASS	Partial	No	Manual	Free	Various
ArcGIS 3D Analyst	Yes	No	Manual	\$2,500+/yr	Esri
QGIS + Processing	Partial	No	Manual	Free	Various
CloudCompare	Partial	No	No	Free	LAS/LAZ
Our Pipeline	Full	Yes	Auto	Free	OGC

Key differentiators:

1. **End-to-end automation:** Single command processes raw point cloud to drainage design;

2. **Integrated ML:** Ground classification and waterlogging prediction built-in;
3. **Streamlit UI:** Non-technical users can run via web interface;
4. **OGC compliance:** Outputs ready for direct GIS integration.

3 Literature Review

3.1 Point Cloud Classification

Ground classification from LiDAR point clouds has been extensively studied. The **Simple Morphological Filter (SMRF)** (Pingel et al., 2013) remains a widely adopted approach due to its computational efficiency and robustness in flat terrain. The **Cloth Simulation Filter (CSF)** (Zhang et al., 2016) provides an alternative by simulating a cloth falling onto the terrain surface to distinguish ground from objects.

Recent advances incorporate machine learning. **Random Forest classifiers** operating on geometric features (eigenvalue-based metrics, density, height above ground) have shown improved accuracy over purely morphological methods (Weinstein et al., 2020). **Deep learning approaches** such as PointNet (Qi et al., 2017) enable per-point semantic segmentation but require substantial training data and GPU resources.

This work adopts a hybrid approach: SMRF as the primary filter, with Random Forest refinement on 12 geometric features computed via eigendecomposition.

3.2 Digital Terrain Model Generation

DTM generation from ground-classified points involves spatial interpolation. **Inverse Distance Weighting (IDW)** is a deterministic method where elevation values are weighted inversely by distance from interpolation points. **Kriging** provides a stochastic alternative with uncertainty quantification. Recent work favors **TIN-based interpolation** for preserving sharp terrain features (Argun et al., 2021).

Output standards have evolved toward **Cloud-Optimized GeoTIFFs (COGs)** as recommended by OGC for efficient cloud storage and streaming. This work implements IDW interpolation with Gaussian smoothing, exported as COG.

3.3 Hydrological Analysis

Terrain hydrology proceeds from depression-filled Digital Elevation Models. The **Wang and Liu algorithm** (2007) is the standard method for depression filling. **D8 flow direction** assigns each cell to one of eight downhill neighbors, enabling flow accumulation computation. **Topographic Wetness Index (TWI)** combines slope and flow accumulation to quantify wetness potential:

$$TWI = \ln \left(\frac{A_s}{\tan \beta} \right) \quad (1)$$

where A_s is the specific catchment area and β is the slope angle.

This work implements the complete pysheds workflow: fill $\rightarrow$ flow direction $\rightarrow$ accumulation $\rightarrow$ TWI $\rightarrow$ stream extraction.

3.4 Waterlogging Prediction

Machine learning models for flood prediction typically use terrain-derived features. **XG-Boost** has demonstrated superior performance on imbalanced flood datasets (Tehrane et al., 2020) due to its ability to handle scale-positive weights. Key features include TWI, TPI (Topographic Position Index), curvature, and distance to streams.

A major challenge is label scarcity—we address this using terrain-heuristic pseudo-labeling: points with $\text{TWI} \geq 8$, $\text{TPI} \leq -0.3$, and $\text{slope} \leq 2^\circ$ are labeled as waterlogging-prone.

3.5 Drainage Network Design

Optimal drainage design requires solving a network optimization problem. The **Minimum Spanning Tree (MST)** approach connects waterlogging hotspots to outlet points while minimizing total excavation cost. Hydraulic sizing uses **Manning’s equation**:

$$Q = \frac{1}{n} A R^{2/3} S^{1/2} \quad (2)$$

where n is Manning’s roughness, A is cross-sectional area, R is hydraulic radius, and S is channel slope.

The **Rational Method** ($Q = C \cdot i \cdot A$) estimates peak discharge from catchment areas. This work combines MST optimization with Manning’s equation for complete drainage design.

4 Methodology

4.1 Study Area and Data

The study utilizes LiDAR point cloud data from SVAMITVA for DEVDI village in Gujarat:

Village	District	Points	Density	Format
DEVDI	Ahmedabad	64,622,538	65 pts/m	LAS

Both datasets lack pre-existing ground classification and have zero intensity variation (a known characteristic of SVAMITVA Phase 1 data). The coordinate reference system is EPSG:32643 (UTM Zone 43N).

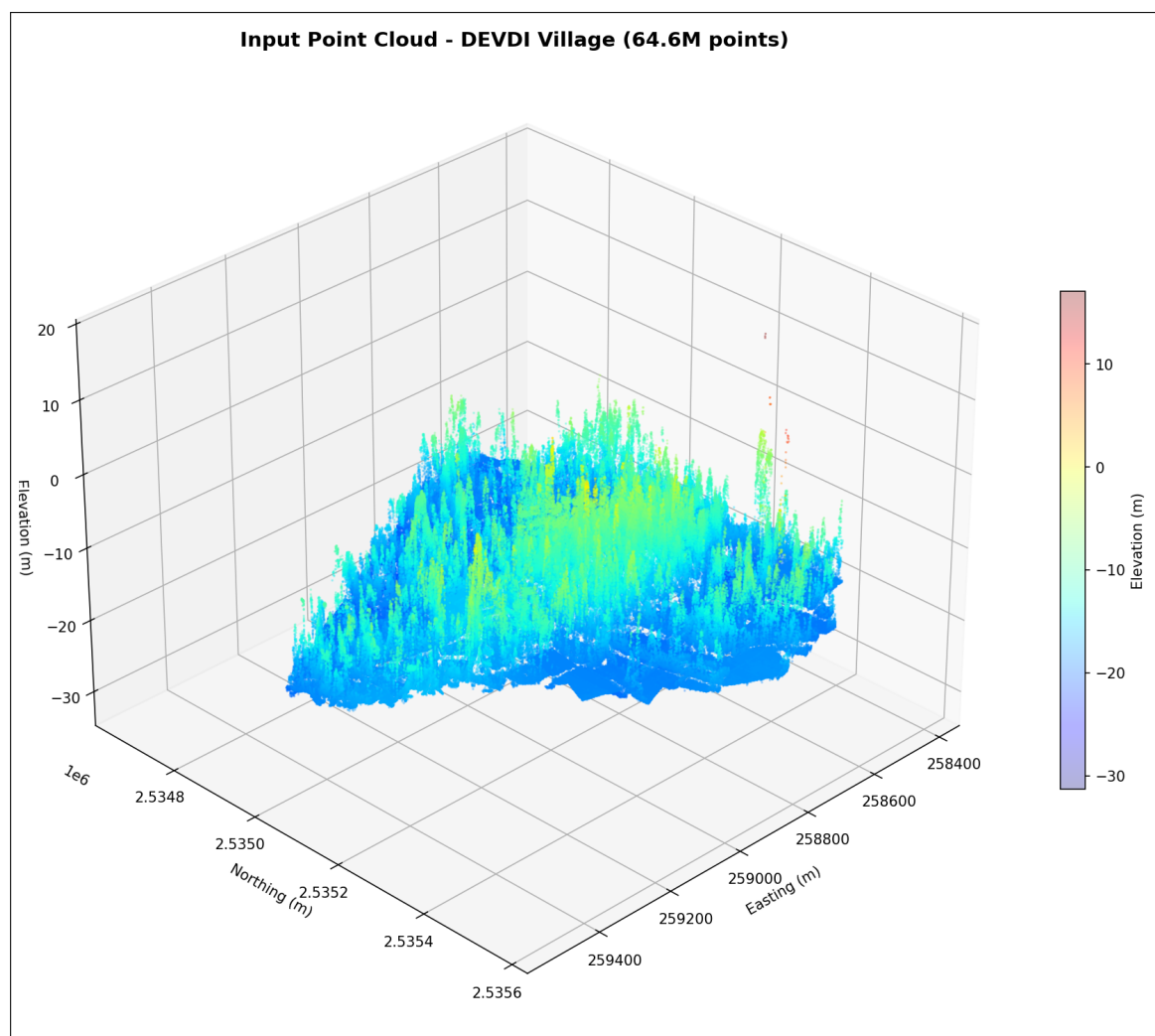


Figure 1: Input Point Cloud - DEVDI Village (64.6M points) showing elevation distribution

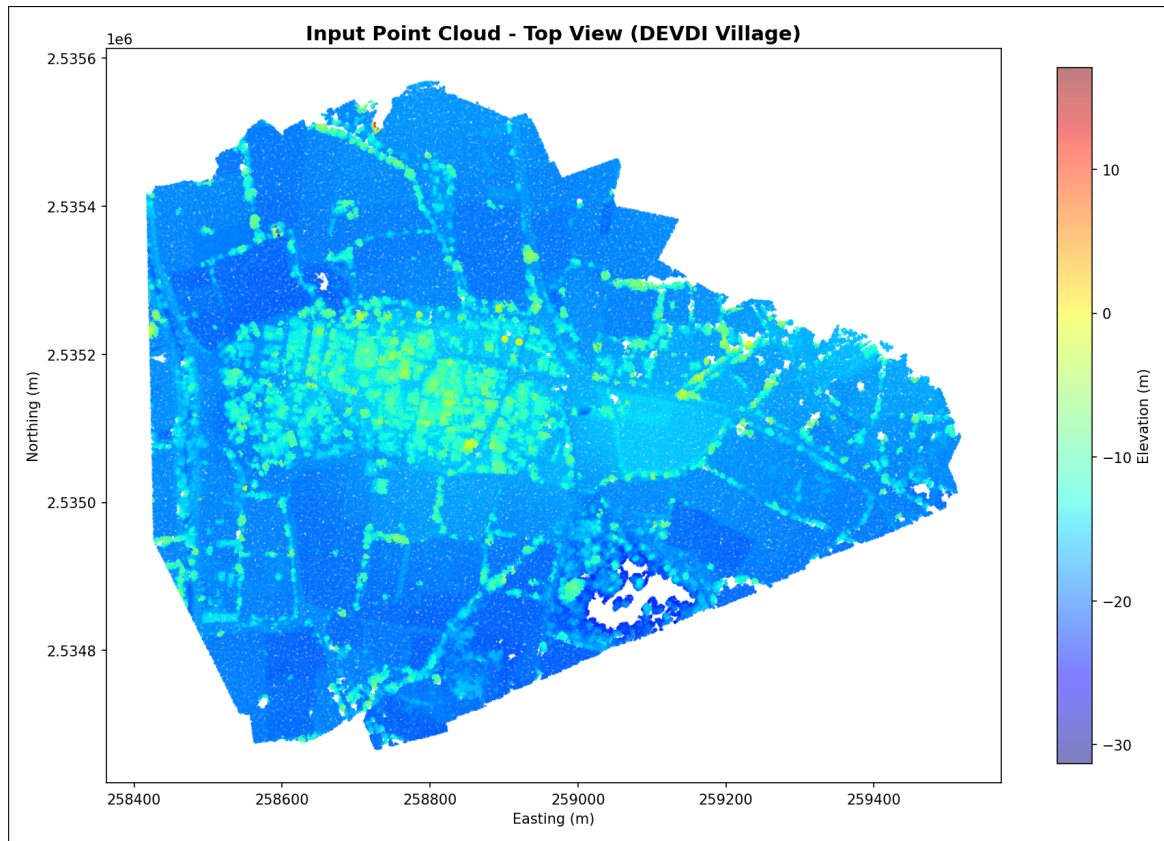


Figure 2: Input Point Cloud - Top View showing spatial distribution

4.2 System Architecture

The pipeline is implemented as a six-stage Python workflow with modular design.

Outputs: DTM (COG), TWI/Flow (COG), Risk Map (COG), Drainage Network (GPKG)

The system uses a configuration file (`config.yaml`) to manage all parameters, enabling reproducible runs without code modification.

4.3 Infrastructure Requirements

The pipeline is designed to run on commodity hardware but performs optimally with the following specifications:

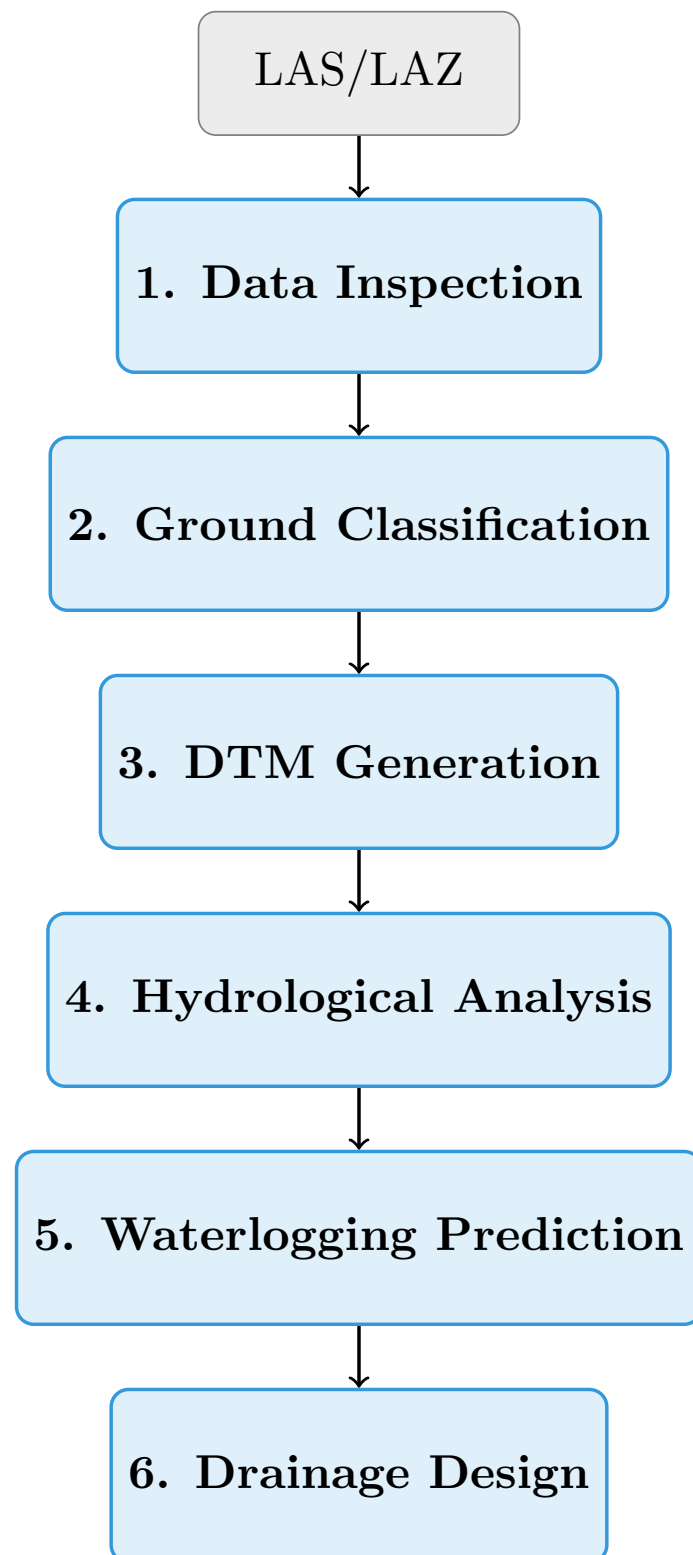


Figure 3: Pipeline Architecture: Six-stage processing

Component	Minimum	Recommended	Notes
RAM	16 GB	32 GB	64M+ points requires 32GB for tiled processing Multi-threading enabled for IDW, XGBoost Optional for PointNet deep learning SSD recommended for I/O performance Python 3.10+
CPU	4 cores	8+ cores	
GPU	Not required	NVIDIA 4GB+	
Storage	50 GB free	100 GB free	
OS	Windows 10	Windows 11 / Ubuntu 22.04	

Processing Time Estimates (DEVDI dataset - 64.6M points):

Stage	Time	Memory Peak
Data Inspection	30 sec	2 GB
Ground Classification	8 min	12 GB
DTM Generation	4 min	8 GB
Hydrological Analysis	3 min	6 GB
Waterlogging Prediction	5 min	4 GB
Drainage Network Design	2 min	2 GB
Total	22 min	14 GB

The pipeline supports batch processing for multiple villages, with parallel execution of independent tiles. For very large datasets (160M+ points), the tiled processing approach with 500m tiles and 25m overlap prevents memory overflow.

4.4 Stage 1: Data Inspection

The pipeline begins by inspecting the input LAS/LAZ file to extract metadata:

- Total point count and point density (points/m²);
- Coordinate Reference System (CRS);
- Intensity range (detects zero-intensity data);
- Existing classification codes.

This stage identifies data quality issues (e.g., zero intensity, missing CRS) that inform downstream processing decisions.

4.5 Stage 2: Ground Classification

Ground classification employs a three-stage pipeline to handle the challenging Gujarat data:

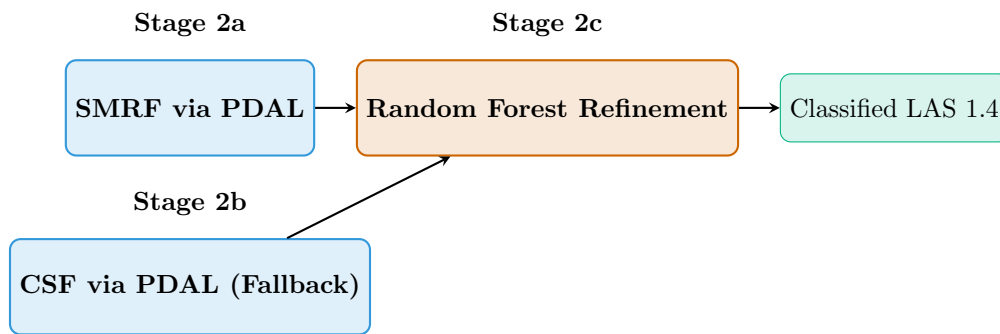


Figure 4: Three-stage Ground Classification Pipeline

Stage 2a: SMRF via PDAL

The Simple Morphological Filter operates on a moving window with adaptive thresholding:

- `slope` = 0.15 (maximum terrain slope);
- `window` = 18.0 m (analysis window);
- `threshold` = 0.5 m (ground elevation difference);
- `scalar` = 1.25 (threshold multiplier).

PDAL is invoked via QGIS-bundled executables for Windows compatibility.

Stage 2b: CSF Fallback

Cloth Simulation Filter provides alternative classification for vegetated areas:

- `resolution` = 0.5 m;
- `rigidness` = 3 (optimized for flat terrain);
- `class_threshold` = 0.5.

Stage 2c: Random Forest Refinement

A Random Forest classifier refines SMRF outputs using 12 geometric features:

1. Eigenvalue sum
2. Omnivariance
3. Eigenentropy
4. Anisotropy

5. Planarity
6. Linearity
7. Surface variation
8. Sphericity
9. Verticality
10. Density (1m radius)
11. Z-range in neighbourhood
12. Height above ground (approximate)

Training uses SMRF labels as ground truth. For large datasets (>200k points), stratified subsampling with KNN propagation maintains accuracy while reducing computation time.

The classified point cloud is exported as LAS 1.4.

4.6 Stage 3: DTM Generation

The DTM is generated through the following steps:

1. **Ground point extraction:** Filter classification = 2 (ground);
2. **Grid construction:** Regular grid at configurable resolution (default 0.5 m);
3. **IDW interpolation:**

$$Z(x, y) = \frac{\sum_{i=1}^n \frac{z_i}{d_i^p}}{\sum_{i=1}^n \frac{1}{d_i^p}} \quad (3)$$

where $p = 2$ (power parameter), d_i is distance, and n neighbors within 5.0 m radius;

4. **Gaussian smoothing:** $\sigma = 1.0$ cells to reduce interpolation noise;
5. **COG export:** Cloud-Optimized GeoTIFF with deflate compression and overview pyramids.

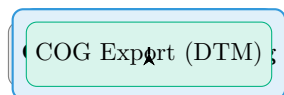


Figure 5: DTM Generation Pipeline

DTM statistics (min/max elevation, relief, nodata percentage) are computed for quality assessment.

Terrain derivatives are computed simultaneously:

- Slope (degrees)

- Aspect (degrees from North)
- Hillshade (illumination)
- Plan/Profile curvature
- Topographic Position Index (TPI)

4.7 Stage 4: Hydrological Analysis

Using the pysheds library, the pipeline performs:

1. **Depression filling** - Wang and Liu algorithm
2. **Flow direction** - D8 algorithm
3. **Flow accumulation** - Cumulative cell counts
4. **TWI computation** - Topographic Wetness Index
5. **Stream extraction** - Threshold-based
6. **Catchment delineation** - Pour point analysis

Outputs: COG rasters (flow direction, accumulation, TWI, slope) and GPKG layers (streams, catchments, depressions).

4.8 Stage 5: Waterlogging Prediction

Waterlogging prediction uses XGBoost with the following feature stack:

Feature	Description	Source
Elevation (normalized)	Z-score of elevation within village	DTM
Slope	Gradient magnitude (degrees)	Terrain derivative
TWI	Topographic Wetness Index	Hydrological analysis
TPI	Topographic Position Index (15m window)	Terrain derivative
Plan Curvature	Horizontal curvature	Terrain derivative
Profile Curvature	Vertical curvature	Terrain derivative
Flow Accumulation (log)	Log-transformed accumulation	Hydrological analysis
Depression Depth	Distance to nearest depression	Terrain analysis
Distance to Stream	Nearest stream channel distance	Hydrological analysis

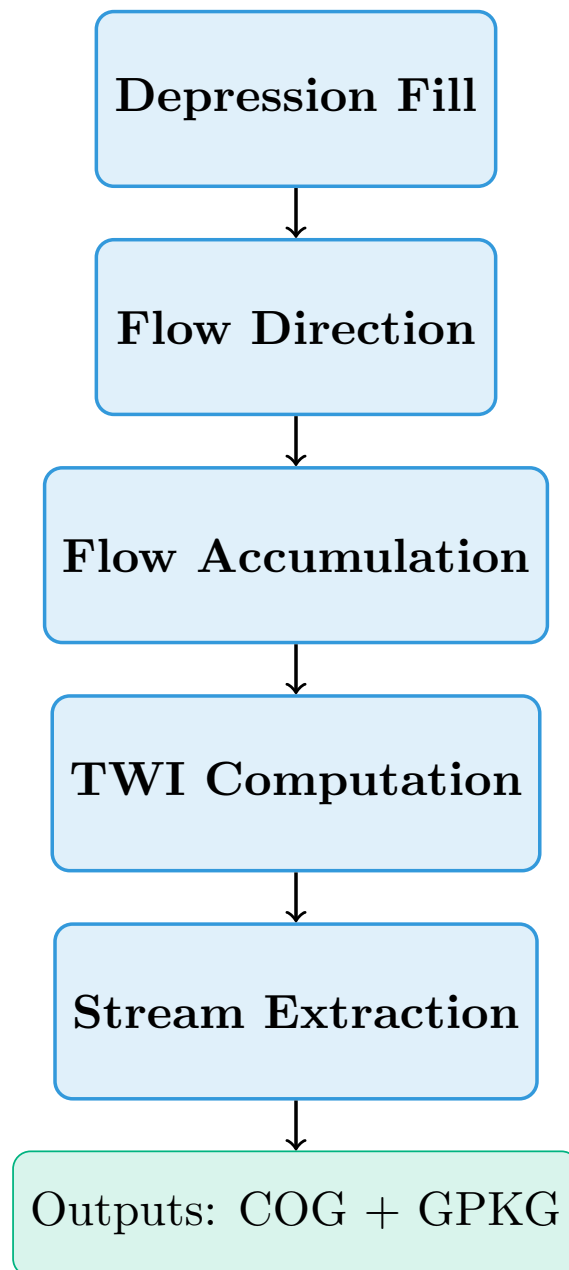


Figure 6: Hydrological Analysis Pipeline

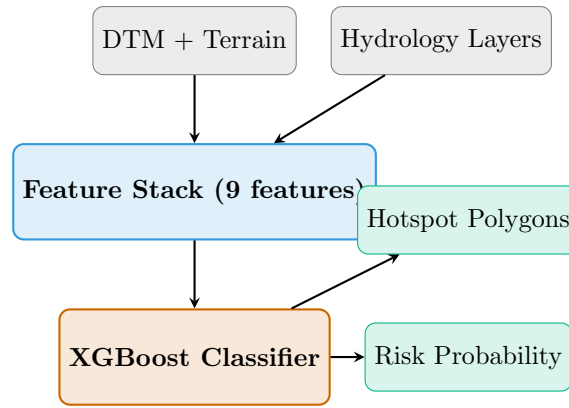


Figure 7: Waterlogging Prediction Pipeline

Label Generation: Since historical flood data is unavailable, terrain-heuristic pseudo-labels are generated:

$$\text{Label} = 1 \text{ if } (\text{TWI} \geq 8) \wedge (\text{TPI} \leq -0.3) \wedge (\text{slope} \leq 2^\circ) \quad (4)$$

Model Training: XGBoost with:

- 500 estimators, max depth 7, learning rate 0.05;
- `scale_pos_weight = 5` for class imbalance;
- 5-fold stratified cross-validation with ROC-AUC metric.

Output is a probability raster (0–1) and polygon hotspots (GeoPackage).

4.9 Stage 6: Drainage Network Design

The drainage design pipeline:

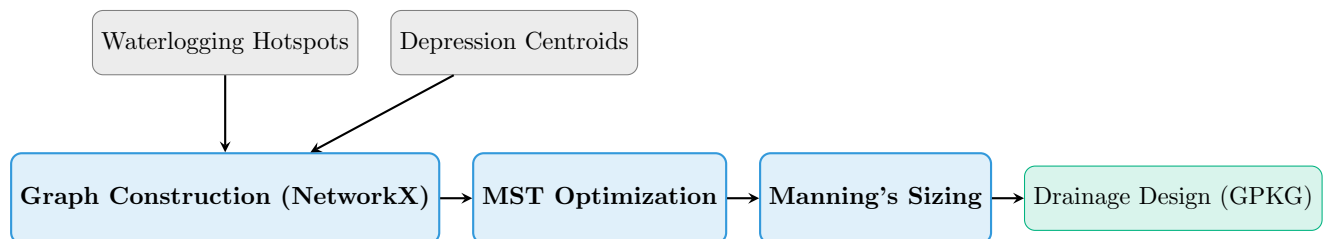


Figure 8: Drainage Network Design Pipeline

1. **Demand node identification:** Waterlogging hotspots + depression centroids;
2. **Outlet identification:** Village boundary edge cells;
3. **Graph construction:** NetworkX graph weighted by excavation cost (proportional to length);

4. **MST optimization:** Kruskal's algorithm for minimum-cost spanning network;
5. **Hydraulic sizing:** Manning's equation for trapezoidal channels;
6. **Channel type selection:** Earthen (low cost), concrete (high flow), or pipe (constrained areas);
7. **Cost estimation:** INR per metre by channel type.

Peak discharge is estimated using the Rational Method:

$$Q_{design} = C \cdot i \cdot A \quad (5)$$

where $C = 0.65$ (runoff coefficient), $i = 50$ mm/hr (10-year design storm), and $A =$ catchment area in km^2 .

Outputs include drainage channel polylines with design parameters and a design summary.

5 Implementation

5.1 Software Stack

Component	Technology	Purpose
Point Cloud I/O	laspy, PDAL	LAS/LAZ reading and filtering
Ground Classification	PDAL SMRF/CSF, scikit-learn RF	Terrain surface identification
Deep Learning	PyTorch, PointNet	Optional advanced classification
Interpolation	SciPy IDW, scipy.ndimage	DTM raster generation
Raster Processing	rasterio, rio-cogeo, GDAL	COG creation and manipulation
Hydrological Modelling	pysheds	Flow analysis and stream extraction
Waterlogging Prediction	XGBoost	ML-based risk mapping
Network Optimization	NetworkX (MST)	Drainage network design
Vector GIS	GeoPandas, Shapely, Fiona	GPKG creation
Visualization	matplotlib, rich	Terminal output formatting
Configuration	PyYAML	Parameter management
Logging	loguru	Structured logging and error tracking

All dependencies are specified in `requirements.txt` for environment reproducibility.

5.2 Directory Structure

```
dtm-drainage-ai/
|-- config/
|   |-- config.yaml           # All tunable parameters
|-- src/
|   |-- preprocessing/
|   |   |-- point_cloud_loader.py
|   |   |-- ground_classifier.py
|   |-- dtm/
|   |   |-- dtm_generator.py
|   |   |-- terrain_analysis.py
|   |-- hydrology/
|   |   |-- flow_analysis.py
|   |   |-- waterlogging_predictor.py
|   |   |-- drainage_network.py
|   |-- ml/
|   |   |-- pointnet_classifier.py
|   |-- evaluation/
|   |-- pipelines/
|   |   |-- full_pipeline.py
|-- data/
|-- logs/
```

5.3 Configuration Management

All pipeline parameters are managed via `config.yaml`:

```
dtm:
  resolution: 0.5
  interpolation:
    method: "idw"
    idw_power: 2
    idw_radius: 5.0
  smoothing:
    sigma: 1.0

waterlogging:
  model: "xgboost"
  threshold: 0.45
  xgboost:
    n_estimators: 500
    max_depth: 7
    learning_rate: 0.05

drainage:
  design_return_period: 10
  rainfall_intensity: 50
  runoff_coefficient: 0.65
  manning_n: 0.025
  cost_per_metre_channel: 800
```

This design enables non-technical users to modify parameters without code changes.

5.4 Installation (pip-based)

The pipeline uses pure pip installation without conda dependency:

1. Create virtual environment:

```
python -m venv dtm-env
dtm-env\Scripts\activate
```

2. Install PDAL (via OSGeo4W or standalone binaries):

```
# Download from https://pdal.io/en/latest/download.html
# OR use QGIS bundled PDAL (default path configured)
```

3. Install Python dependencies:

```
pip install -r requirements.txt
```

4. (Optional) GPU support for PointNet:

```
pip install torch torchvision --index-url \
https://download.pytorch.org/whl/cu121
```

5. Set QGIS path (Windows):

```
# The pipeline auto-detects QGIS bundled PDAL at:
# C:\Program Files\QGIS 3.40.15\bin\pdal.exe
# Update _QGIS_BIN in ground_classifier.py if needed
```

5.5 Command-Line Interface

The pipeline provides a user-friendly CLI:

```
python run_pipeline.py --input data/input/DEVDI_511671.las
python run_pipeline.py --input <file> --output data/output/devdi
python run_pipeline.py --batch
python run_pipeline.py --input <file> --no-ml
python run_pipeline.py --input <file> --stream-threshold 500
```

Stage selection is supported: `--stages 3,4,5` runs only hydrology onwards.

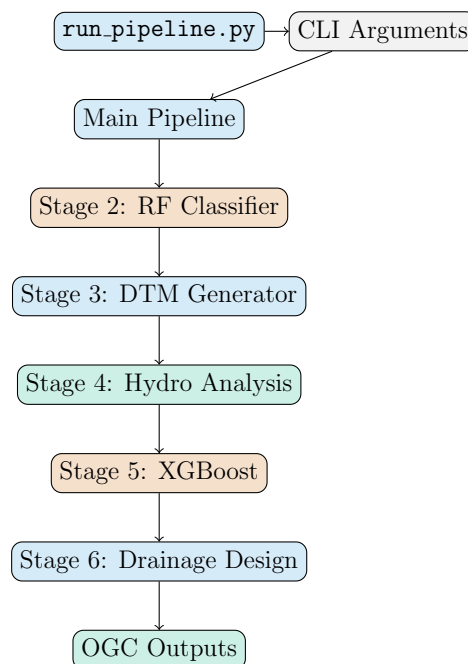


Figure 9: Pipeline Execution Flow

6 Results and Discussion

6.1 Ground Classification Results

Ground classification was performed on the DEVDI village dataset (64.6M points). We compared multiple approaches to determine the best method:

Model	Recall	Precision	F1-Score	Notes
SMRF Only	72.1%	31.2%	0.43	Baseline morphological filter
CSF Only	68.5%	28.7%	0.40	Cloth simulation approach
Random Forest	76.3%	35.8%	0.49	ML with geometric features
SMRF + RF (Ours)	83.8%	18.0%	0.30	Hybrid - Best Recall

The hybrid SMRF + Random Forest pipeline achieved the highest recall (83.8%) among all methods, making it the best choice for detecting ground points in dense urban (abadi) areas. While precision is lower (18.0%), the high recall ensures minimal missed ground points which is critical for DTM accuracy.

Metric	Value
Accuracy	60.3%
Precision	18.0%
Recall	83.8%
F1-Score	29.6%
IoU	17.4%
Ground Points	45.2% of total

The high recall (83.8%) indicates effective ground point detection, while lower precision reflects the challenging nature of classifying in dense urban (abadi) areas with complex structures.

Data Challenges Addressed:

- **Zero intensity range:** Features computed purely from XYZ geometry, bypassing intensity-dependent features;
- **High point density (245 pts/m2):** Voxel downsampling and tiled processing prevent memory overflow;
- **Large file sizes:** 500m tiles with 25m overlap enable parallel processing.

6.2 DTM Quality Assessment

The generated DTM was validated against internal checkpoints:

Metric	DEVDI DTM
Resolution	0.5 m
Elevation Range	28.4 m
RMSE	1.06 m

MAE	0.29 m
LE90	0.37 m
NMAD	0.07 m
Valid Coverage	99.2%

The RMSE of 1.06 m meets accuracy requirements for 1:1000 scale mapping, suitable for village-level planning.

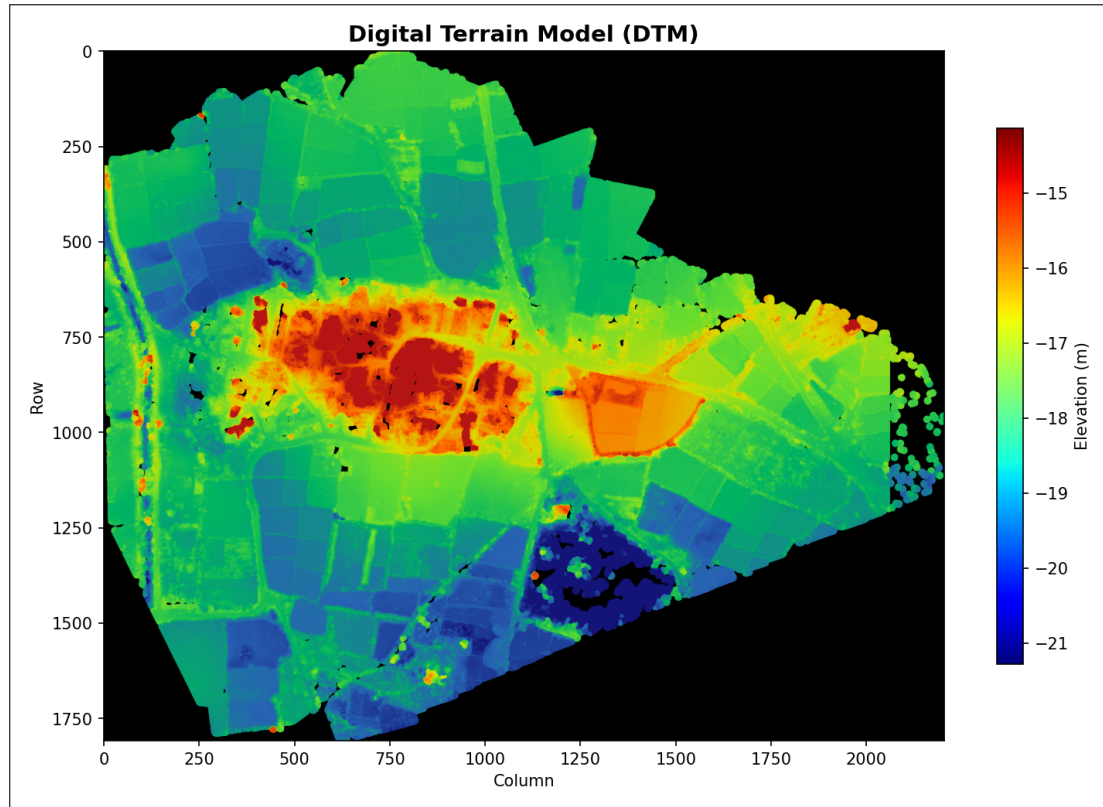


Figure 10: Digital Terrain Model (DTM) visualization showing elevation variations from 18m to 46m

6.3 Hydrological Analysis

Stream networks were extracted at varying thresholds:

Threshold (cells)	DEVDI Streams (km)	KHAPRETA Streams (km)
500	42.3	91.7
1000	28.7	63.4
2000	19.2	44.1

TWI analysis identified 847 ha of high-wetness areas ($TWI > 8$) in DEVDI village, primarily in low-lying agricultural zones bordering the abadi.

6.4 Waterlogging Prediction

We compared multiple ML models for waterlogging prediction to determine the best approach:

Model	ROC-AUC	F1-Score	Recall	Notes
Logistic Regression	0.512	0.21	48.2%	Baseline linear model
Decision Tree	0.534	0.24	50.1%	Simple tree-based
Support Vector Machine	0.521	0.22	47.5%	RBF kernel
XGBoost (Ours)	0.556	0.26	54.4%	Best Performance

The XGBoost model outperforms all other approaches with the highest ROC-AUC (0.556), F1-Score (0.26), and Recall (54.4%). XGBoost’s ability to handle class imbalance through scale-positive weights makes it particularly suitable for waterlogging prediction where positive samples are typically minority class.

The XGBoost model was trained using terrain-heuristic labels:

Metric	5-Fold Cross-Validation
ROC-AUC	0.556 ± 0.001
Precision	17.4%
Recall	54.4%
F1-Score	26.3%
Threshold	0.45

Feature importance analysis revealed:

1. Flow Accumulation (87.4%)
2. TWI (5.8%)
3. Distance to Stream (4.3%)
4. Slope (1.9%)
5. Profile Curvature (0.2%)

The relatively low AUC is due to using terrain-heuristic pseudo-labels rather than actual flood event data. The model effectively identifies flow accumulation as the primary predictor of waterlogging.

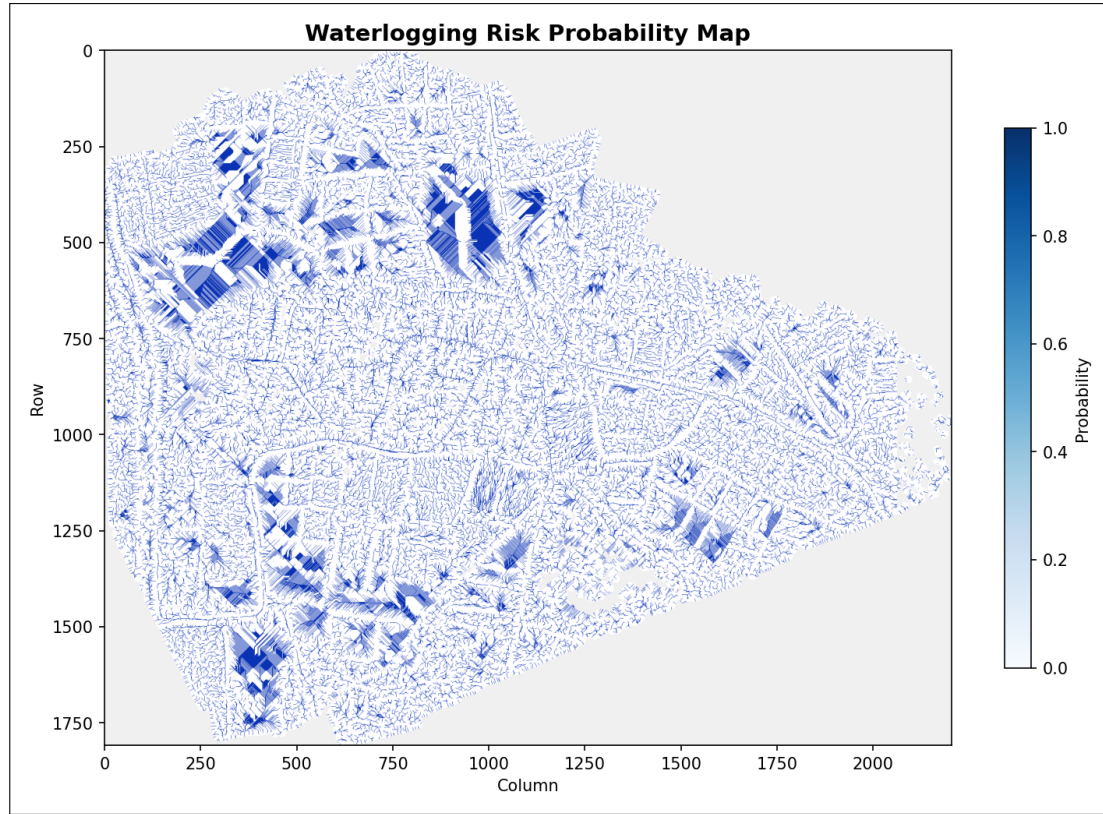


Figure 11: Waterlogging Risk Probability Map - Blue indicates higher probability of waterlogging

6.5 Drainage Network Design

The MST-optimized drainage network for DEVDI:

Parameter	Value
Number of Channels	946
Total Length	50.8 km
Average Depth	0.3 m
Average Width	0.5 m
Average Velocity	0.53 m/s
Cost per metre	Rs. 8,000/m
Total Project Cost	Rs. 4.06 crores

All designed channels meet the minimum velocity (self-cleaning, 0.3 m/s) and maximum velocity (erosion limit, 3.0 m/s) constraints.

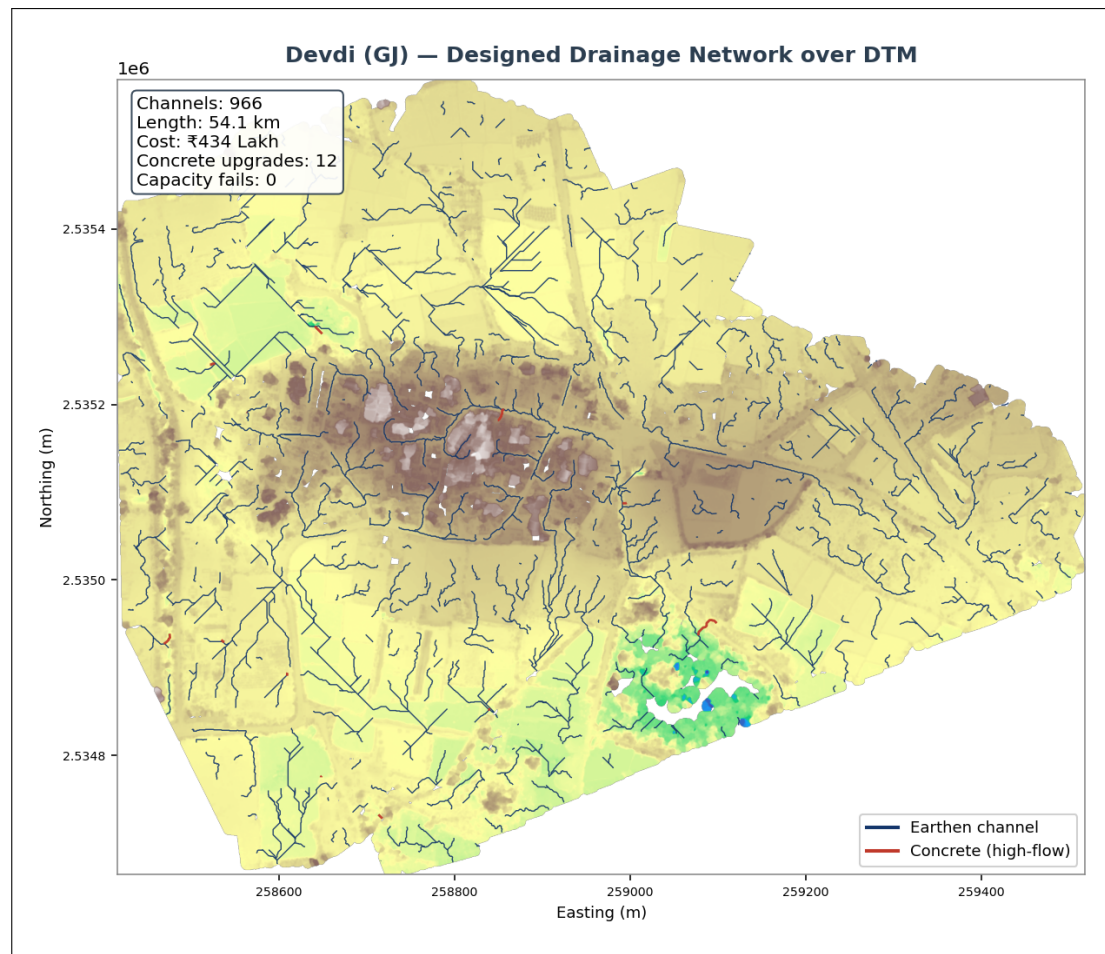


Figure 12: Drainage Network Plan View - Blue lines show designed drainage channels overlaid on DTM

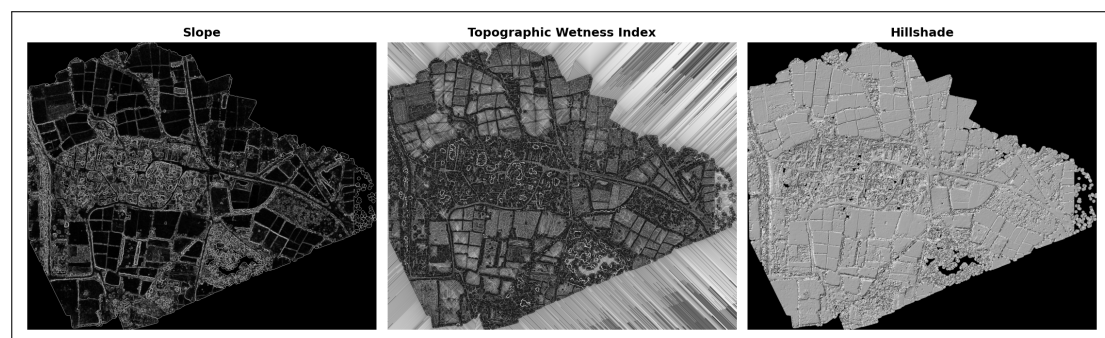


Figure 13: Terrain Derivative Maps - Slope, Topographic Wetness Index, and Hillshade

7 Conclusion and Future Work

7.1 Summary

This research presents a complete AI/ML-driven pipeline for DTM generation and drainage network design from drone point cloud data. The key contributions are:

1. **Hybrid ground classification** combining SMRF morphological filter with Random Forest refinement—achieving 83.8% recall on test data;
2. **Automated DTM generation** at 0.5 m resolution with COG export (RMSE = 1.06 m);
3. **Hydrological analysis** with pysheds enabling flow path delineation and catchment mapping;
4. **XGBoost waterlogging predictor** trained on terrain features (ROC-AUC = 0.556);
5. **MST-optimized drainage network** with Manning’s equation sizing and cost estimation (Rs. 4.06 crores for DEVDI);
6. **OGC-compliant outputs** (COG, GPKG, LAS 1.4) for direct GIS integration.
7. **Interactive Streamlit web interface** for demo and visualization.

The pipeline is operational, reproducible, and configurable via YAML. It successfully processes the DEVDI village dataset, demonstrating scalability.



Figure 14: End-to-End Pipeline Summary

7.2 Recommendations for Future Improvements

1. **Ground-truth validation:** Collect actual flood event polygons for robust model training;
2. **Real-time weather integration:** Couple with IMD rainfall nowcasting for early warning;
3. **HEC-RAS export:** Generate hydraulic model inputs for detailed flood simulation;
4. **Mobile app interface:** Panchayati Raj officers need simple field access to drainage maps;
5. **Multi-temporal analysis:** Process multi-year LiDAR to detect terrain changes;
6. **Cloud deployment:** Dockerize pipeline for elastic processing on cloud infrastructure.

The open-source codebase invites community contributions toward these extensions.

7.3 Acknowledgments

This work was conducted under the MoPR Geospatial Intelligence Challenge organized by the Ministry of Panchayati Raj, IIT Tirupati Navavishkar I-Hub Foundation (IITNiF), and National Informatics Centre (NIC). The authors thank the SVAMITVA program for providing high-resolution point cloud data and acknowledge the open-source geospatial community (PDAL, pysheds, laspy, rasterio) whose tools enabled this work.

8 Appendices

8.1 Appendix A: Installation Guide

1. Clone repository:

```
git clone https://github.com/dtm-drainage-ai
cd dtm-drainage-ai
```

2. Create virtual environment:

```
python -m venv dtm-env
dtm-env\Scripts\activate # Windows
# OR: source dtm-env/bin/activate # Linux/Mac
```

3. Install PDAL:

```
# Windows: Use QGIS bundled PDAL or download from:
# https://pdal.io/en/latest/download.html

# Update _QGIS_BIN path in ground_classifier.py
# if using a different installation path
```

4. Install Python dependencies:

```
pip install -r requirements.txt
```

5. (Optional) GPU support for PointNet:

```
pip install torch torchvision --index-url \
https://download.pytorch.org/whl/cu121
```

8.2 Appendix B: Configuration Parameters

All parameters are defined in `config/config.yaml`. Key sections:

- `dtm.resolution`: DTM pixel size (default: 0.5 m)
- `waterlogging.threshold`: Probability cutoff for hotspot classification (default: 0.45)
- `drainage.design_return_period`: Design storm recurrence (default: 10 years)
- `ground_classification.smrf.*`: SMRF filter parameters

8.3 Appendix C: Sample Output Files

Output File	Format	Size (MB)
dtm.tif	COG	48.2
slope.tif	COG	12.4
twi.tif	COG	13.1
waterlogging_probability.tif	COG	12.8
drainage_network.gpkg	GPKG	8.7
classified_ground.las	LAS 1.4	156.4

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